

CHAPTER 4

LIVE LOADS

4.1 DEFINITIONS

The following definitions apply to the provisions of this chapter.

FIXED LADDER: A ladder that is permanently attached to a structure, building, or equipment.

GRAB BAR SYSTEM: A bar and associated anchorages and attachments to the structural system, for the support of body weight in locations such as toilets, showers, and tub enclosures.

GUARDRAIL SYSTEM: A system of components, including anchorages and attachments to the structural system, near open sides of an elevated surface for the purpose of minimizing the possibility of a fall from the elevated surface by people, equipment, or material.

HANDRAIL SYSTEM: A rail grasped by hand for guidance and support and associated anchorages and attachments to the structural system.

HELIPAD: A structural surface that is used for landing, taking off, taxiing, and parking of helicopters.

LIVE LOAD: A load produced by the use and occupancy of the building or other structure that does not include construction or environmental loads, such as wind load, snow load, rain load, earthquake load, flood load, or dead load.

ROOF LIVE LOAD: A load on a roof produced (1) during maintenance by workers, equipment, and materials, and (2) during the life of the structure by movable objects, such as planters or other similar small decorative appurtenances that are not occupancy related. An occupancy-related live load on a roof such as rooftop assembly areas, rooftop decks, and vegetative or landscaped roofs with occupiable areas, is considered to be a live load rather than a roof live load.

SCREEN ENCLOSURE: A building or part thereof, in whole or in part self-supporting, having walls and a roof of insect or sun screening using fiberglass, aluminum, plastic, or similar lightweight netting material, which encloses an occupancy or use such as outdoor swimming pools, patios or decks, and horticultural and agricultural production facilities.

VEHICLE BARRIER SYSTEM: A system of components, including anchorages and attachments to the structural system near open sides or walls of garage floors or ramps, that acts as a restraint for vehicles.

4.2 LOADS NOT SPECIFIED

For occupancies or uses not designated in this chapter, the live load shall be determined in accordance with a method approved by the Authority Having Jurisdiction.

4.3 UNIFORMLY DISTRIBUTED LIVE LOADS

4.3.1 Required Live Loads. The live loads used in the design of buildings and other structures shall be the maximum loads expected by the intended use or occupancy but shall in no case be

less than the minimum uniformly distributed unit loads required by Table 4.3-1.

4.3.2 Provision for Partitions. In office buildings and in other buildings where partition locations are subject to change, provisions for partition weight shall be made, whether or not partitions are shown on the plans. The partition load shall not be less than 15 psf (0.72 kN/m²).

EXCEPTION: A partition live load is not required where the minimum specified live load is 80 psf (3.83 kN/m²) or greater.

4.3.3 Partial Loading. The full intensity of the appropriately reduced live load applied only to a portion of a structure or member shall be accounted for if it produces a more unfavorable load effect than the same intensity applied over the full structure or member. Roof live loads shall be distributed as specified in Table 4.3-1.

4.4 CONCENTRATED LIVE LOADS

Floors, roofs, and other similar surfaces shall be designed to support the uniformly distributed live loads prescribed in Section 4.3 or the concentrated load, in pounds or kilonewtons (kN), given in Table 4.3-1, whichever produces the greater load effects. Unless otherwise specified, the indicated concentration shall be assumed to be uniformly distributed over an area 2.5 ft (762 mm) by 2.5 ft (762 mm) and shall be located so as to produce the maximum load effects in the members.

4.5 LOADS ON HANDRAIL, GUARDRAIL, GRAB BAR, AND VEHICLE BARRIER SYSTEMS, AND ON FIXED LADDERS

4.5.1 Handrail and Guardrail Systems. Handrail and guardrail systems shall be designed to resist a single concentrated load of 200 lb (0.89 kN) applied in any direction at any point on the handrail or top rail to produce the maximum load effect on the element being considered and to transfer this load through the supports to the structure.

4.5.1.1 Uniform Load. Handrail and guardrail systems shall also be designed to resist a load of 50 lb/ft (pound-force per linear foot) (0.73 kN/m) applied in any direction along the handrail or top rail and to transfer this load through the supports to the structure. This load need not be assumed to act concurrently with the concentrated load specified in Section 4.5.1.

EXCEPTIONS: The uniform load need not be considered for the following occupancies:

1. one- and two-family dwellings, and
2. factory, industrial, and storage occupancies in areas that are not accessible to the public and that serve an occupant load not greater than 50.

Table 4.3-1 Minimum Uniformly Distributed Live Loads, L_o , and Minimum Concentrated Live Loads

Occupancy or Use	Uniform, L_o psf (kN/m ²)	Live Load Reduction Permitted? (Sec. No.)	Multiple-Story Live Load Reduction Permitted? (Sec. No.)	Concentrated lb (kN)	Also See Section
Apartments (See Residential)					
Access floor systems					
Office use	50 (2.40)	Yes (4.7.2)	Yes (4.7.2)	2,000 (8.90)	
Computer use	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)	2,000 (8.90)	
Armories and drill rooms	150 (7.18)	No (4.7.5)	No (4.7.5)		
Assembly areas					
Fixed seats (fastened to floors)	60 (2.87)	No (4.7.5)	No (4.7.5)		
Lobbies	100 (4.79)	No (4.7.5)	No (4.7.5)		
Movable seats	100 (4.79)	No (4.7.5)	No (4.7.5)		
Platforms (assembly)	100 (4.79)	No (4.7.5)	No (4.7.5)		
Stage floors	150 (7.18)	No (4.7.5)	No (4.7.5)		
Reviewing stands, grandstands, and bleachers	100 (4.79)	No (4.7.5)	No (4.7.5)		4.14
Stadiums and arenas with fixed seats (fastened to the floor)	60 (2.87)	No (4.7.5)	No (4.7.5)		4.14
Other assembly areas	100 (4.79)	No (4.7.5)	No (4.7.5)		
Balconies and decks	1.5 times the live load for the area served. Not required to exceed 100 psf (4.79 kN/m ²)	Yes (4.7.2)	Yes (4.7.2)		
Catwalks for maintenance access	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)	300 (1.33)	
Corridors					
First floor	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)		
Other floors	Same as occupancy served except as indicated				
Dining rooms and restaurants	100 (4.79)	No (4.7.5)	No (4.7.5)		
Dwellings (See Residential)					
Elevator machine room grating (on area of 2 in. by 2 in. (50 mm by 50 mm))		—	—	300 (1.33)	
Finish light floor plate construction (on area of 1 in. by 1 in. (25 mm by 25 mm))		—	—	200 (0.89)	
Fire escapes	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)		
On single-family dwellings only	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)		
Fixed ladders		—	—	See Sec. 4.5.4	
Garages (See Section 4.10)					
Passenger vehicles only	40 (1.92)	No (4.7.4)	Yes (4.7.4)	See Sec. 4.10.1	
Trucks and buses	See Sec. 4.10.2	—	—	See Sec. 4.10.2	
Handrails and Guardrails	See Sec. 4.5.1	—	—	See Sec. 4.5.1	
Grab bars		—	—	See Sec. 4.5.2	
Helipads (See Section 4.11)					
Helicopter takeoff weight 3,000 lb (13.35 kN) or less	40 (1.92)	No (4.11.1)	—	See Sec. 4.11.2	
Helicopter takeoff weight more than 3,000 lb (13.35 kN)	60 (2.87)	No (4.11.1)	—	See Sec. 4.11.2	
Hospitals					
Operating rooms, laboratories	60 (2.87)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Patient rooms	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Corridors above first floor	80 (3.83)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Hotels (See Residential)					
Libraries					
Reading rooms	60 (2.87)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Stack rooms	150 (7.18)	No (4.7.3)	Yes (4.7.3)	1,000 (4.45)	4.13
Corridors above first floor	80 (3.83)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Manufacturing					
Light	125 (6.00)	No (4.7.3)	Yes (4.7.3)	2,000 (8.90)	
Heavy	250 (11.97)	No (4.7.3)	Yes (4.7.3)	3,000 (13.35)	
Office buildings					
File and computer rooms shall be designed for heavier loads based on anticipated occupancy					
Lobbies and first-floor corridors	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)	2,000 (8.90)	
Offices	50 (2.40)	Yes (4.7.2)	Yes (4.7.2)	2,000 (8.90)	
Corridors above first floor	80 (3.83)	Yes (4.7.2)	Yes (4.7.2)	2,000 (8.90)	

continues

Table 4.3-1. (Continued) Minimum Uniformly Distributed Live Loads, L_o , and Minimum Concentrated Live Loads

Occupancy or Use	Uniform, L_o psf (kN/m ²)	Live Load Reduction Permitted? (Sec. No.)	Multiple-Story Live Load Reduction Permitted? (Sec. No.)	Concentrated lb (kN)	Also See Section
Penal institutions					
Cell blocks	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)		
Corridors	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)		
Recreational uses					
Bowling alleys, poolrooms, and similar uses	75 (3.59)	No (4.7.5)	No (4.7.5)		
Dance halls and ballrooms	100 (4.79)	No (4.7.5)	No (4.7.5)		
Gymnasiums	100 (4.79)	No (4.7.5)	No (4.7.5)		
Residential					
One- and two-family dwellings					
Uninhabitable attics without storage	10 (0.48)	Yes (4.7.2)	Yes (4.7.2)		4.12.1
Uninhabitable attics with storage	20 (0.96)	Yes (4.7.2)	Yes (4.7.2)		4.12.2
Habitable attics and sleeping areas	30 (1.44)	Yes (4.7.2)	Yes (4.7.2)		
All other areas except stairs	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)		
All other residential occupancies					
Private rooms and corridors serving them	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)		
Public rooms	100 (4.79)	No (4.7.5)	No (4.7.5)		
Corridors serving public rooms	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)		
Roofs					
Ordinary flat, pitched, and curved roofs	20 (0.96)	Yes (4.8.2)	—		4.8.1
Roof areas used for occupants	Same as occupancy served	Yes (4.8.3)	—		
Roof areas used for assembly purposes	100 (4.70)	Yes (4.8.3)			
Vegetative and landscaped roofs					
Roof areas not intended for occupancy	20 (0.96)	Yes (4.8.2)	—		
Roof areas used for assembly purposes	100 (4.70)	Yes (4.8.3)	—		
Roof areas used for other occupancies	Same as occupancy served	Yes (4.8.3)	—		
Awnings and canopies					
Fabric construction supported by a skeleton structure	5 (0.24)	No (4.8.2)	—		
Screen enclosure support frame	5 (0.24) based on the tributary area of the roof supported by the frame member	No (4.8.2)	—	200 (0.89)	
All other construction	20 (0.96)	Yes (4.8.2)	—		4.8.1
Primary roof members, exposed to a work floor					
Single panel point of lower chord of roof trusses or any point along primary structural members supporting roofs over manufacturing, storage warehouses, and repair garages				2,000 (8.90)	
All other primary roof members		—	—	300 (1.33)	
All roof surfaces subject to maintenance workers		—	—	300 (1.33)	
Schools					
Classrooms	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Corridors above first floor	80 (3.83)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
First-floor corridors	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Scuttles, skylight ribs, and accessible ceilings				200 (0.89)	
Sidewalks, vehicular driveways, and yards subject to trucking	250 (11.97)	No (4.7.3)	Yes (4.7.3)	8,000 (35.60)	4.15
Stairs and exit ways	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)	300 (1.33)	4.16
One- and two-family dwellings only	40 (1.92)	Yes (4.7.2)	Yes (4.7.2)	300 (1.33)	4.16
Storage areas above ceilings	20 (0.96)	Yes (4.7.2)	Yes (4.7.2)		
Storage warehouses (shall be designed for heavier loads if required for anticipated storage)					
Light	125 (6.00)	No (4.7.3)	Yes (4.7.3)		
Heavy	250 (11.97)	No (4.7.3)	Yes (4.7.3)		

continues

Table 4.3-1. (Continued) Minimum Uniformly Distributed Live Loads, L_o , and Minimum Concentrated Live Loads

Occupancy or Use	Uniform, L_o psf (kN/m ²)	Live Load Reduction Permitted? (Sec. No.)	Multiple-Story Live Load Reduction Permitted? (Sec. No.)	Concentrated lb (kN)	Also See Section
Stores					
Retail					
First floor	100 (4.79)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Upper floors	75 (3.59)	Yes (4.7.2)	Yes (4.7.2)	1,000 (4.45)	
Wholesale, all floors	125 (6.00)	No (4.7.3)	Yes (4.7.3)	1,000 (4.45)	
Vehicle barriers				See Sec. 4.5.3	
Walkways and elevated platforms (other than exit ways)	60 (2.87)	Yes (4.7.2)	Yes (4.7.2)		
Yards and terraces, pedestrian	100 (4.79)	No (4.7.5)	No (4.7.5)		

4.5.1.2 Guardrail System Component Loads. Balusters, panel fillers, and guardrail infill components, including all rails except the handrail and the top rail, shall be designed to resist a horizontally applied normal load of 50 lb (0.22 kN) on an area not to exceed 12 in. by 12 in. (305 mm by 305 mm), including openings and space between rails and located so as to produce the maximum load effects. Reactions due to this loading are not required to be superimposed with the loads specified in Sections 4.5.1 and 4.5.1.1.

4.5.2 Grab Bar Systems. Grab bar systems shall be designed to resist a single concentrated load of 250 lb (1.11 kN) applied in any direction at any point on the grab bar to produce the maximum load effect.

4.5.3 Vehicle Barrier Systems. Vehicle barrier systems for passenger vehicles shall be designed to resist a single load of 6,000 lb (26.70 kN) applied horizontally in any direction to the barrier system and shall have anchorages or attachments capable of transferring this load to the structure. For design of the system, the load shall be assumed to act at heights between 1 ft 6 in. (460 mm) and 2 ft 3 in. (686 mm) above the floor or ramp surface, located to produce the maximum load effects. The load shall be applied on an area not to exceed 12 in. by 12 in. (305 mm by 305 mm). This load is not required to act concurrently with any handrail or guardrail system loadings specified in Section 4.5.1. Vehicle barrier systems in garages accommodating trucks and buses shall be designed in accordance with *AASHTO LRFD Bridge Design Specifications*.

4.5.4 Fixed Ladders. Fixed ladders with rungs shall be designed to resist a single concentrated load of 300 lb (1.33 kN) applied at any point to produce the maximum load effect on the element being considered. The number and position of additional concentrated live load units shall be a minimum of 1 unit of 300 lb (1.33 kN) for every 10 ft (3.05 m) of ladder height.

Where rails of fixed ladders extend above a floor or platform at the top of the ladder, each side rail extension shall be designed to resist a single concentrated live load of 100 lb (0.445 kN) applied in any direction at any height up to the top of the side rail extension. Ships ladders with treads instead of rungs shall be designed to resist the stair loads given in Table 4.3-1.

4.6 IMPACT LOADS

4.6.1 General. The live loads specified in Sections 4.3 through 4.5 shall be assumed to include adequate allowance for ordinary

impact conditions. Provision shall be made in the structural design for uses and loads that involve unusual vibration and impact forces.

4.6.2 Elevators. All elements subject to dynamic loads from elevators shall be designed for impact loads and deflection limits prescribed by ASME A17.

4.6.3 Machinery. For the purpose of design, the weight of machinery and moving loads shall be increased as follows to allow for impact: (1) light machinery, shaft- or motor-driven, 20%; and (2) reciprocating machinery or power-driven units, 50%. All percentages shall be increased where specified by the manufacturer.

4.6.4 Elements Supporting Hoists for Façade Access and Building Maintenance Equipment. Structural elements that support hoists for façade and building maintenance equipment shall be designed for a live load of 2.5 times the rated load of the hoist or the stall load of the hoist, whichever is larger.

4.6.5 Fall Arrest and Lifeline Anchorages. Fall arrest and lifeline anchorages and structural elements that support these anchorages shall be designed for a live load of 3,100 lb (13.8 kN) for each attached lifeline in every direction that a fall arrest load may be applied.

4.7 REDUCTION IN UNIFORM LIVE LOADS

4.7.1 General. Except for roof uniform live loads, all other minimum uniformly distributed live loads, L_o in Table 4.3-1, are permitted to be reduced in accordance with the requirements of Sections 4.7.2 through 4.7.6.

4.7.2 Reduction in Uniform Live Loads. Subject to the limitations of Sections 4.7.3 through 4.7.6, members for which a value of $K_{LL}A_T$ is 400 ft² (37.16 m²) or more are permitted to be designed for a reduced live load in accordance with the following formula:

$$L = L_o \left(0.25 + \frac{15}{\sqrt{K_{LL}A_T}} \right) \quad (4.7-1)$$

$$L = L_o \left(0.25 + \frac{4.57}{\sqrt{K_{LL}A_T}} \right) \quad (4.7-1si)$$

Table 4.7-1 Live Load Element Factor, K_{LL}

Element	K_{LL} ^a
Interior columns	4
Exterior columns without cantilever slabs	4
Edge columns with cantilever slabs	3
Corner columns with cantilever slabs	2
Edge beams without cantilever slabs	2
Interior beams	2
All other members not identified, including	1
Edge beams with cantilever slabs	
Cantilever beams	
One-way slabs	
Two-way slabs	
Members without provisions for continuous shear transfer normal to their span	

^aIn lieu of the preceding values, K_{LL} is permitted to be calculated.

where

L = reduced design live load per ft² (m²) of area supported by the member

L_o = unreduced design live load per ft² (m²) of area supported by the member (see Table 4.3-1)

K_{LL} = live load element factor (see Table 4.7-1)

A_T = tributary area in ft² (m²).

L shall not be less than $0.50L_o$ for members supporting one floor, and L shall not be less than $0.40L_o$ for members supporting two or more floors.

4.7.3 Heavy Live Loads. Live loads that exceed 100 lb/ft² (4.79 kN/m²) shall not be reduced.

EXCEPTION: Live loads for members supporting two or more floors are permitted to be reduced by a maximum of 20%, but the reduced live load shall not be less than L as calculated in Section 4.7.2.

4.7.4 Passenger Vehicle Garages. The live loads shall not be reduced in passenger vehicle garages.

EXCEPTION: Live loads for members supporting two or more floors are permitted to be reduced by a maximum of 20%, but the reduced live load shall not be less than L as calculated in Section 4.7.2.

4.7.5 Assembly Uses. Live loads shall not be reduced in assembly uses.

4.7.6 Limitations on One-Way Slabs. The tributary area, A_T , for one-way slabs shall not exceed an area defined by the slab span times a width normal to the span of 1.5 times the slab span.

4.8 REDUCTION IN ROOF LIVE LOADS

4.8.1 General. The minimum uniformly distributed roof live loads, L_o in Table 4.3-1, are permitted to be reduced in accordance with the requirements of Sections 4.8.2 and 4.8.3.

Where uniform roof live loads are reduced to less than 20 lb/ft² (0.96 kN/m²) in accordance with Section 4.8.2 and are applied to the design of structural members arranged so as to create continuity, the reduced roof live load shall be applied to adjacent spans or to alternate spans, whichever produces the greatest unfavorable load effect.

4.8.2 Ordinary Roofs, Awnings, and Canopies. Ordinary flat, pitched, and curved roofs, and awning and canopies other than

those of fabric construction supported by a skeleton structure, are permitted to be designed for a reduced roof live load, as specified in Eq. (4.8-1), or other controlling combinations of loads, as specified in Chapter 2, whichever produces the greater load effect. In structures such as greenhouses, where special scaffolding is used as a work surface for workers and materials during maintenance and repair operations, a lower roof load than specified in Eq. (4.8-1) shall not be used unless approved by the Authority Having Jurisdiction. On such structures, the minimum roof live load shall be 12 psf (0.58 kN/m²).

$$L_r = L_o R_1 R_2 \quad \text{where} \quad 12 \leq L_r \leq 20 \quad (4.8-1)$$

$$L_r = L_o R_1 R_2 \quad \text{where} \quad 0.58 \leq L_r \leq 0.96 \quad (4.8-1si)$$

where

L_r = reduced roof live load per ft² (m²) of horizontal projection supported by the member and

L_o = unreduced design roof live load per ft² (m²) of horizontal projection supported by the member (see Table 4.3-1).

The reduction factors R_1 and R_2 shall be determined as follows:

$$R_1 = \begin{matrix} 1 & \text{for } A_T \leq 200 \text{ ft}^2 \\ 1.2 - 0.001A_T & \text{for } 200 \text{ ft}^2 < A_T < 600 \text{ ft}^2 \\ 0.6 & \text{for } A_T \geq 600 \text{ ft}^2 \end{matrix}$$

in SI:

$$R_1 = \begin{matrix} 1 & \text{for } A_T \leq 18.58 \text{ m}^2 \\ 1.2 - 0.011A_T & \text{for } 18.58 \text{ m}^2 < A_T < 55.74 \text{ m}^2 \\ 0.6 & \text{for } A_T \geq 55.74 \text{ m}^2 \end{matrix}$$

where A_T = tributary area in ft² (m²) supported by the member and

$$R_2 = \begin{matrix} 1 & \text{for } F \leq 4 \\ 1.2 - 0.05F & \text{for } 4 < F < 12 \\ 0.6 & \text{for } F \geq 12 \end{matrix}$$

where, for a pitched roof, F = number of inches of rise per foot (in SI: $F = 0.12 \times \text{slope}$, with slope expressed in percentage points) and, for an arch or dome, F = rise-to-span ratio multiplied by 32.

4.8.3 Occupiable Roofs. Roofs that have an occupancy function, such as roof gardens or other special purposes, are permitted to have their uniformly distributed live load reduced in accordance with the requirements of Section 4.7.

Roofs used for other special purposes shall be designed for appropriate loads as approved by the Authority Having Jurisdiction.

4.9 CRANE LOADS

4.9.1 General. The crane live load shall be the rated capacity of the crane. Design loads for the runway beams, including connections and support brackets, of moving bridge cranes and monorail cranes shall include the maximum wheel loads

of the crane and the vertical impact, lateral, and longitudinal forces induced by the moving crane.

4.9.2 Maximum Wheel Load. The maximum wheel loads shall be the wheel loads produced by the weight of the bridge, as applicable, plus the sum of the rated capacity and the weight of the trolley with the trolley positioned on its runway at the location where the resulting load effect is maximized.

4.9.3 Vertical Impact Force. The maximum wheel loads of the crane determined in accordance with Section 4.9.2 shall be increased by the percentages shown in the following text to account for the effects of vertical impact or vibration:

Monorail cranes (powered)	25
Cab-operated or remotely operated bridge cranes (powered)	25
Pendant-operated bridge cranes (powered)	10
Bridge cranes or monorail cranes with hand-gear bridge, trolley, and hoist	0

4.9.4 Lateral Force. The lateral force on crane runway beams with electrically powered trolleys shall be calculated as 20% of the sum of the rated capacity of the crane and the weight of the hoist and trolley. The lateral force shall be assumed to act horizontally at the traction surface of a runway beam, in either direction perpendicular to the beam, and shall be distributed with due regard to the lateral stiffness of the runway beam and supporting structure.

4.9.5 Longitudinal Force. The longitudinal force on crane runway beams, except for bridge cranes with hand-gear bridges, shall be calculated as 10% of the maximum wheel loads of the crane. The longitudinal force shall be assumed to act horizontally at the traction surface of a runway beam in either direction parallel to the beam.

4.10 GARAGE LOADS

4.10.1 Passenger Vehicle Garages. Floors in garages or portions of a building used for the storage of motor vehicles shall be designed for the uniformly distributed live loads of Table 4.3-1 or the following concentrated load: (1) for garages restricted to passenger vehicles accommodating not more than nine passengers, 3,000 lb (13.35 kN) acting on an area of 4.5 in. by 4.5 in. (114 mm by 114 mm); and (2) for mechanical parking structures without slab or deck that are used for storing passenger vehicles only, 2,250 lb (10 kN) per wheel.

4.10.2 Truck and Bus Garages. Live loads in garages or portions of a building used for the storage of trucks and buses shall be in accordance with *AASHTO LRFD Bridge Design Specifications*; however, provisions for fatigue and dynamic load allowance therein are not required to be applied.

4.11 HELIPAD LOADS

4.11.1 General. The live loads shall not be reduced. The labeling of helicopter capacity shall be as required by the Authority Having Jurisdiction.

4.11.2 Concentrated Helicopter Loads. Two single concentrated loads, 8 ft (2.44 m) apart, shall be applied on the landing area (representing the helicopter's two main landing gear, whether skid type or wheeled type), each having a magnitude of 0.75 times the maximum takeoff weight of the helicopter and located to produce the maximum load effect on the structural elements under consideration. The concentrated loads

shall be applied over an area of 8 in. by 8 in. (200 mm by 200 mm) and are not required to act concurrently with other uniform or concentrated live loads.

A single concentrated load of 3,000 lb (13.35 kN) shall be applied over an area of 4.5 in. by 4.5 in. (114 mm by 114 mm), located so as to produce the maximum load effects on the structural elements under consideration. The concentrated load is not required to act concurrently with other uniform or concentrated live loads.

4.12 UNINHABITABLE ATTICS

4.12.1 Uninhabitable Attics without Storage. In residential occupancies, uninhabitable attic areas without storage are those where the maximum clear height between the joist and rafter is less than 42 in. (1,067 mm) or where there are not two or more adjacent trusses with web configurations capable of accommodating an assumed rectangle 42 in. (1,067 mm) in height by 24 in. (610 mm) in width, or greater, within the plane of the trusses. The live load in Table 4.3-1 need not be assumed to act concurrently with any other live load requirement.

4.12.2 Uninhabitable Attics with Storage. In residential occupancies, uninhabitable attic areas with storage are those where the maximum clear height between the joist and rafter is 42 in. (1,067 mm) or greater or where there are two or more adjacent trusses with web configurations capable of accommodating an assumed rectangle 42 in. (1,067 mm) in height by 24 in. (610 mm) in width, or greater, within the plane of the trusses. For attics constructed of trusses, the live load in Table 4.3-1 need only be applied to those portions of the bottom chords where both of the following conditions are met:

- The attic area is accessible from an opening not less than 20 in. (508 mm) in width by 30 in. (762 mm) in length that is located where the clear height in the attic is a minimum of 30 in. (762 mm); and
- The slope of the truss bottom chord is no greater than 2 units vertical to 12 units horizontal (9.5% slope).

The remaining portions of the bottom chords shall be designed for a uniformly distributed concurrent live load of not less than 10 lb/ft² (0.48 kN/m²).

4.13 LIBRARY STACK ROOMS

The live loading for library stack rooms provided in Table 4.3-1 applies to stack room floors that support nonmobile, double-faced library book stacks subject to the following limitations:

- The nominal book stack unit height shall not exceed 90 in. (2,290 mm);
- The nominal shelf depth shall not exceed 12 in. (305 mm) for each face;
- Parallel rows of double-faced book stacks shall be separated by aisles not less than 36 in. (914 mm) wide.

4.14 SEATING FOR ASSEMBLY USES

In addition to the vertical live loads provided in Table 4.3-1 for reviewing stands, grandstands, and bleachers, and for stadiums and arenas with fixed seats fastened to the floor, the design shall include horizontal swaying forces applied to each row of the seats as follows:

1. 24 lb per linear ft (0.35 kN/m) of seat applied in a direction parallel to each row of seats and
2. 10 lb per linear ft (0.15 kN/m) of seat applied in a direction perpendicular to each row of seats.

The parallel and perpendicular horizontal swaying forces need not be applied simultaneously.

4.15 SIDEWALKS, VEHICULAR DRIVEWAYS, AND YARDS SUBJECT TO TRUCKING

4.15.1 Uniform Loads. Uniform loads, other than that provided in Table 4.3-1, shall also be considered where appropriate, in accordance with an approved method which contains provisions for truck loadings.

4.15.2 Concentrated Loads. The concentrated wheel load provided in Table 4.3-1 shall be applied on an area of 4.5 in. by 4.5 in. (114 mm by 114 mm).

4.16 STAIR TREADS

The concentrated load on stair treads provided in Table 4.3-1 for stair and exit ways, and for stairs in one- and two-family dwellings, shall be applied on an area of 2 in. by 2 in. (50 mm by 50 mm) and is to be applied nonconcurrently with the uniform load.

4.17 SOLAR PANEL LOADS

4.17.1 Roof Loads at Solar Panels. Roof structures that support solar panel systems shall be designed to resist each of the following conditions:

1. The uniform and concentrated roof live loads specified in Table 4.3-1 with the solar panel system dead loads.

EXCEPTION: The roof live load need not be applied to the area covered by solar panels where the clear space between the panels and the roof surface is 24 in. (610 mm) or less.

2. The uniform and concentrated roof live loads specified in Table 4.3-1 without the solar panel system present.

4.17.2 Load Combination. Roof systems that provide support for solar panel systems shall be designed for the load combinations specified in Chapter 2.

4.17.3 Open-Grid Roof Structures Supporting Solar Panels. Structures with open-grid framing and no roof deck or sheathing supporting solar panel systems shall be designed to support the uniform and concentrated roof live loads specified in Table 4.3-1, except that the uniform roof live load shall be permitted to be reduced to 12 psf (0.57 kN/m²).

4.18 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

This section lists the consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter:

AASHTO LRFD Bridge Design Specifications, 7th Ed., American Association of State Highway and Transportation Officials, 2014, with 2015 interim revisions

Cited in: Section 4.5.3, Section 4.10.2

ASME A17, American National Standard Safety Code for Elevators and Escalators, American Society of Mechanical Engineers, 2013

Cited in: Section 4.6.2

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